

datison^e
Training Courses

Git@Work!

Daryoosh Hoorra



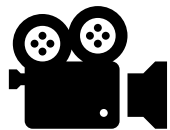
دوره آموزش کاربرد Git

کد دوره: DTS.DEV.GIT.WRK.01

نگارش: ۱.۰

مدرس: داریوش هورا

تاریخ: تابستان ۱۴۰۵



dati**one**

انتشار این محتوای آموزشی با ذکر منبع بلامانع می باشد

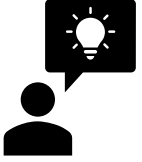




git



FOR
DUMMIES



سخن بزرگان

“ To git, or not to git,
that is the question. ”

William Shakespeare



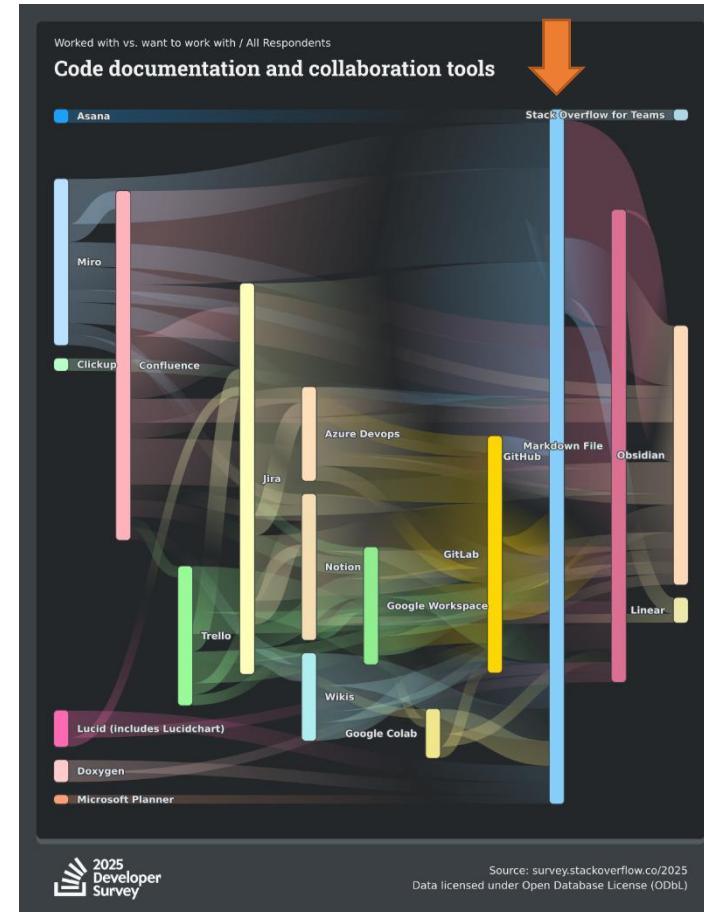
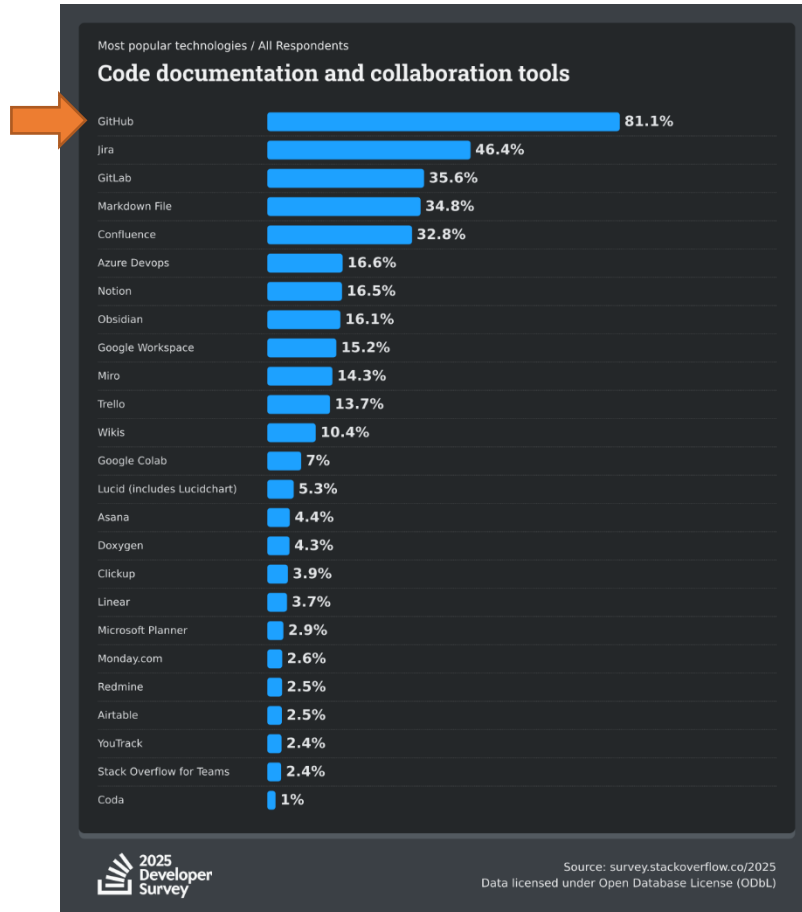
That question is: Do you love, or hate the Git?



2025 Stack Overflow Developer Survey

<https://survey.stackoverflow.co/2025>

They loves Git!





چرا Git؟

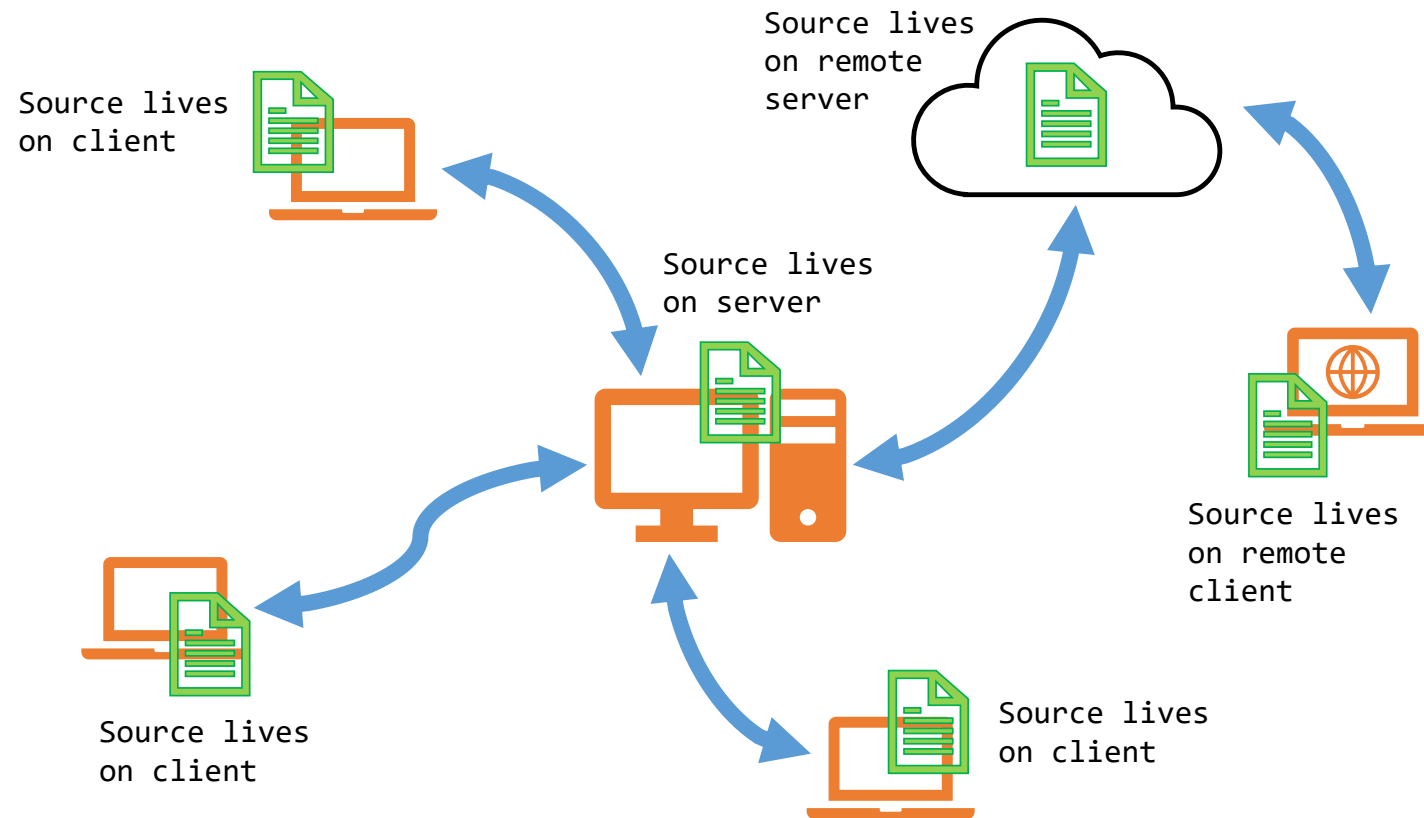
- Easy to use, Git is simple!
- Free and open-source.
- Cross platform, running on Windows, macOS and of course Linux, where Git was borne.
- Lightening and lightweight, so it's high performance.
- Integrated with all most development tools (IDEs).
- Widely utilized in most DevOps frameworks.
- Classy! Everyone wants you to know about that.



چرا Git؟

- Distributed, so it's reliable. (important!)
- Reliable because it's distributed. (very important!)

Everything is Everywhere



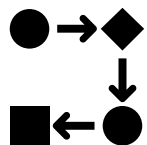


PracticeTime



مخاطبان دوره

- Project's team members
- Artifact creators
- Everyone who create, update and manage files



دوره های تکمیلی

همنیاز دوره
null •

پیشنیاز دوره
null •



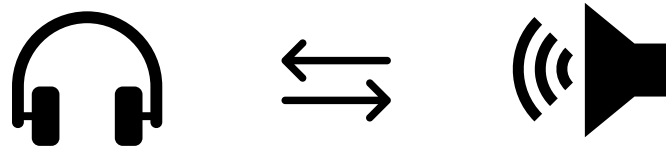
منابع

- Git
 - <https://git-scm.com/>
- Pro Git Book, Apress pub.
 - <https://git-scm.com/book/en/v2>
- W3Schools
 - <https://www.w3schools.com/git/default.asp>



توضیحات

- تمامی اسکرین شات های این آموزش از محیط ویندوز ۱۱ تهیه شده، در سایر نگارشهای ویندوز و سیستمهای عامل لینوکس و مک ممکن است تفاوت ظاهری وجود داشته باشد.
- در تاریخ انتشار این آموزش، نگارش مورد استفاده گیت 2.x می باشد.
- برای اجرای دستورات این آموزش لازم است اتصال به اینترنت برقرار باشد.
- وضوح و بلندی صدای ویدیو!



daryoosh@hoora.info



@datis.one



@datisone



<https://github.com/daryoosh-hoora>

Repositories: **Git-at-Work**



عناوین دوره

- Session 1 - Git Internal

- Concepts
- Definitions
- Skeleton
- Installation

- Session 2 – Git External

- Configuration
- Basics
- Working scenarios

فصل اول

انتشار این محتوای آموزشی با ذکر منبع بلامانع می باشد

Previously on Git!

- Why Git?
- Who are audiences?
- What are references?

What is Git?



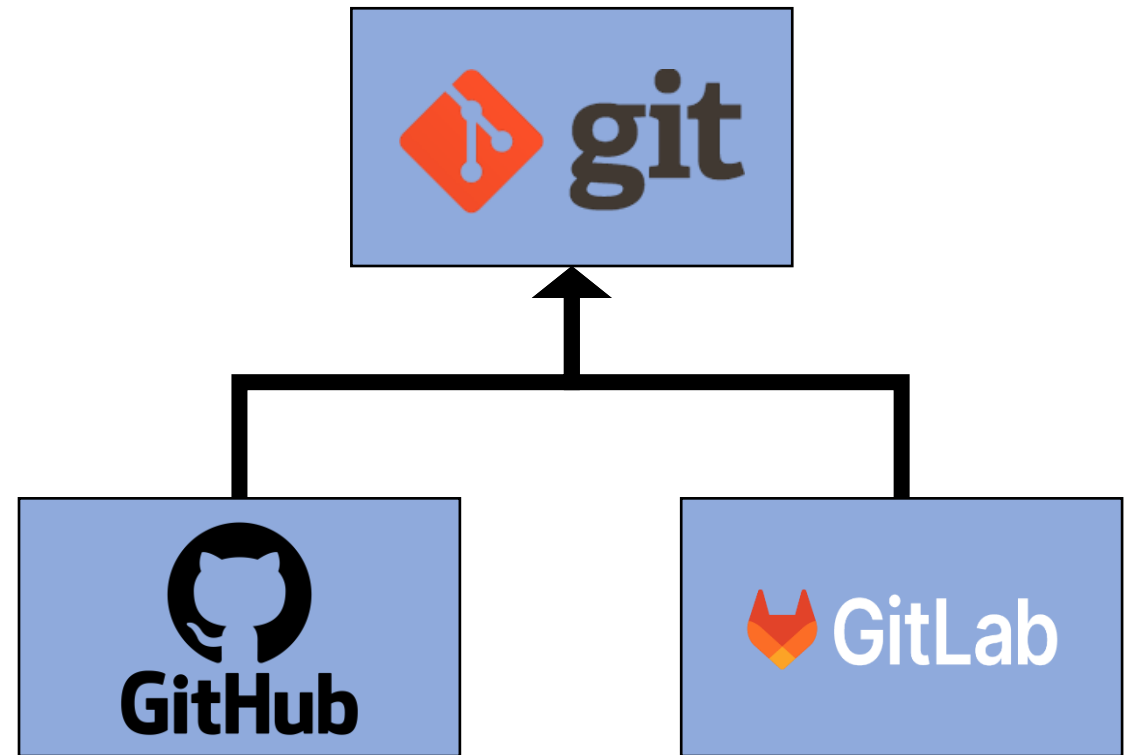
Git is a **free and open source** distributed version control system designed to handle everything from small to very large projects with speed and efficiency.

Git is **lightning fast** and has a huge ecosystem of **GUIs, hosting services, and command-line tools**.

<https://git-scm.com/>

What is NOT?

- Git is NOT a file storage system
- Git is NOT GitHub
- Git is NOT GitLab

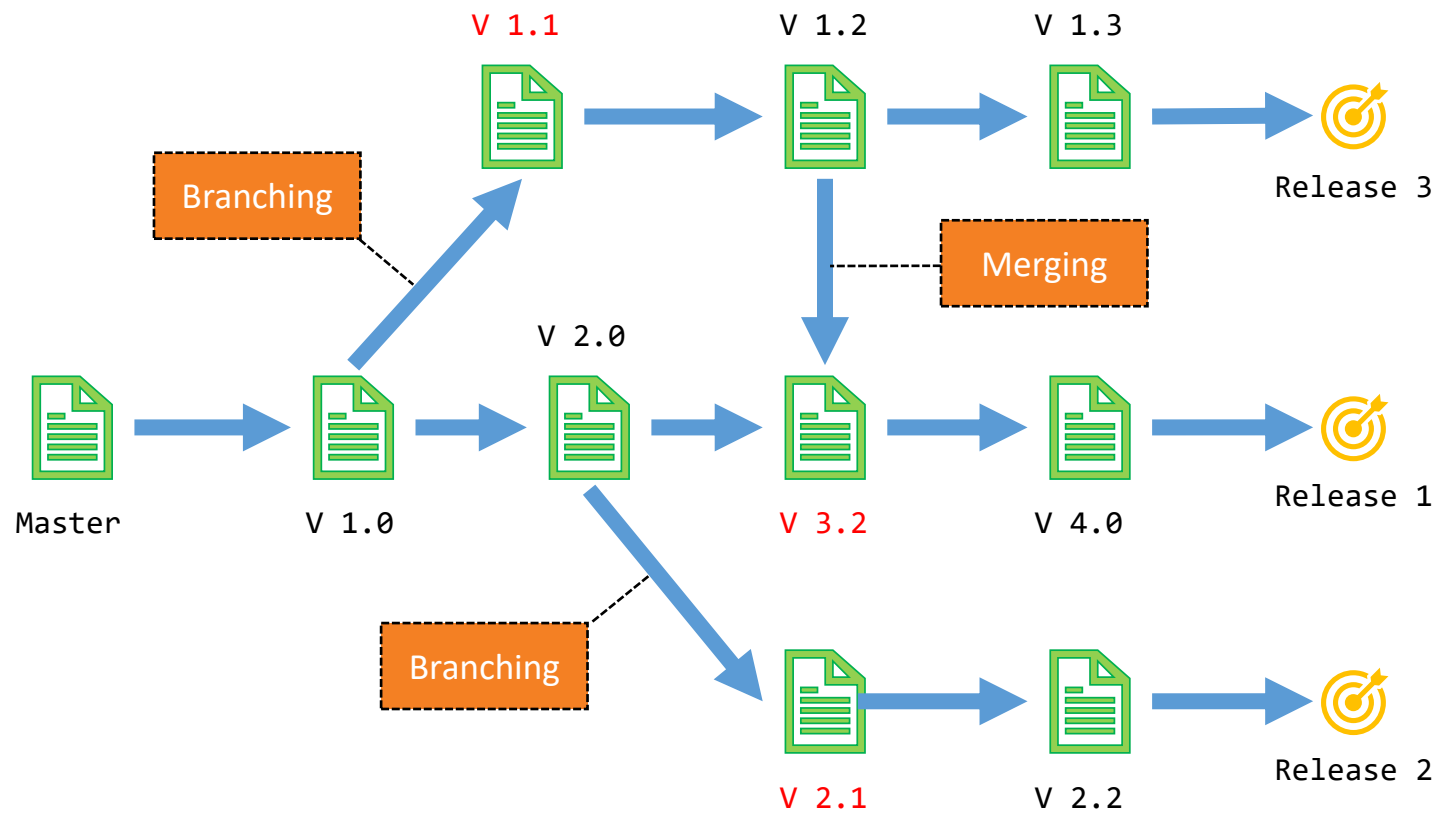


History

- Linux (OS)
- Community (People)
- BitKeeper (VCS)
- Linus Torvalds (Person)
- 2005 (Birthdate)

What is VCS

- Version control is a system that **records changes** to a file or set of files **over time** so that you can **recall** specific versions later.
 - Any type of files
 - Revert back to previous state
 - Compare changes over time
 - Who makes changes and when
 - etc.



Types of VCS

- Stupid version control approach
 - Copy files into another folder!
 - Time-stamped folder for clever people.
- Confusing about folders
- No change tracking
- Client crash down

Types of VCS

- Local Version Control Systems
 - A simple database lives on client
- + Changes tracking
- Teamwork impossible
- Client crash down

Types of VCS

- Centralized Version Control Systems (CVCSs)
 - A shared database on server
 - You may know SVN
- + Teamwork collaboration is possible
- + Administrative controls
- Server crash down

Types of VCS

- Distributed Version Control Systems (DVCSs)  So, this is Git
 - --everything-is-local
 - Clients do not check out the latest changes, they fully mirror the database, including its all history
- + Collaborate with different group of people in different workflows within same project
- + Server failover by restoring database from each clients
- + Avoid network latency overhead
- + It is hard to do something that is not undoable or losing data

Who needs Git?

- A lonely alone programmer, someone who developing solo.
- Software development teams to collaborate on codebase.
- Non-programmers, to work on their documents.
- Git is not suitable for archiving images, movies and any other large files.

Ecosystem

- GUIs: <https://git-scm.com/downloads/guis>
- Hosting services: <https://git-scm.com/tools/hosting>
- CLI tools: <https://git-scm.com/tools/command-line>
- Integration extensions

Workout

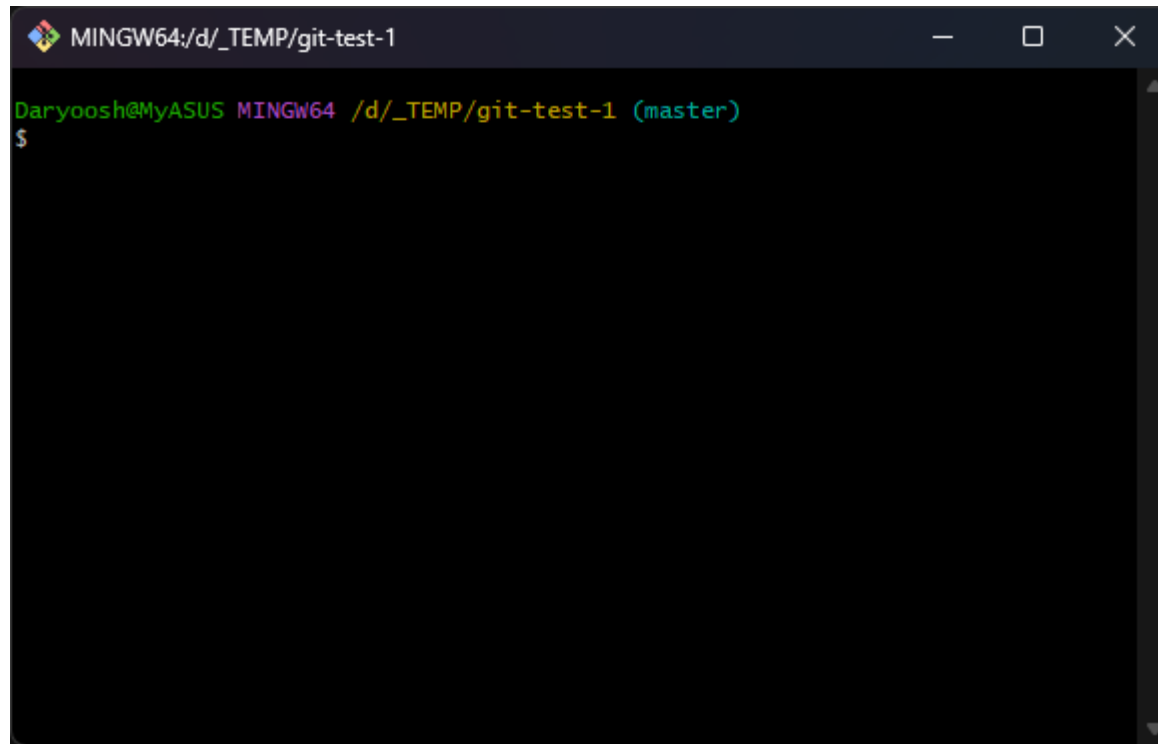
- Create your own [GitHub](#) account
- Create your first repository

CLI vs. GUI

- CLI is native
- CLI is generic
- CLI is only place to run all commands
- CLI is foundation of GUIs

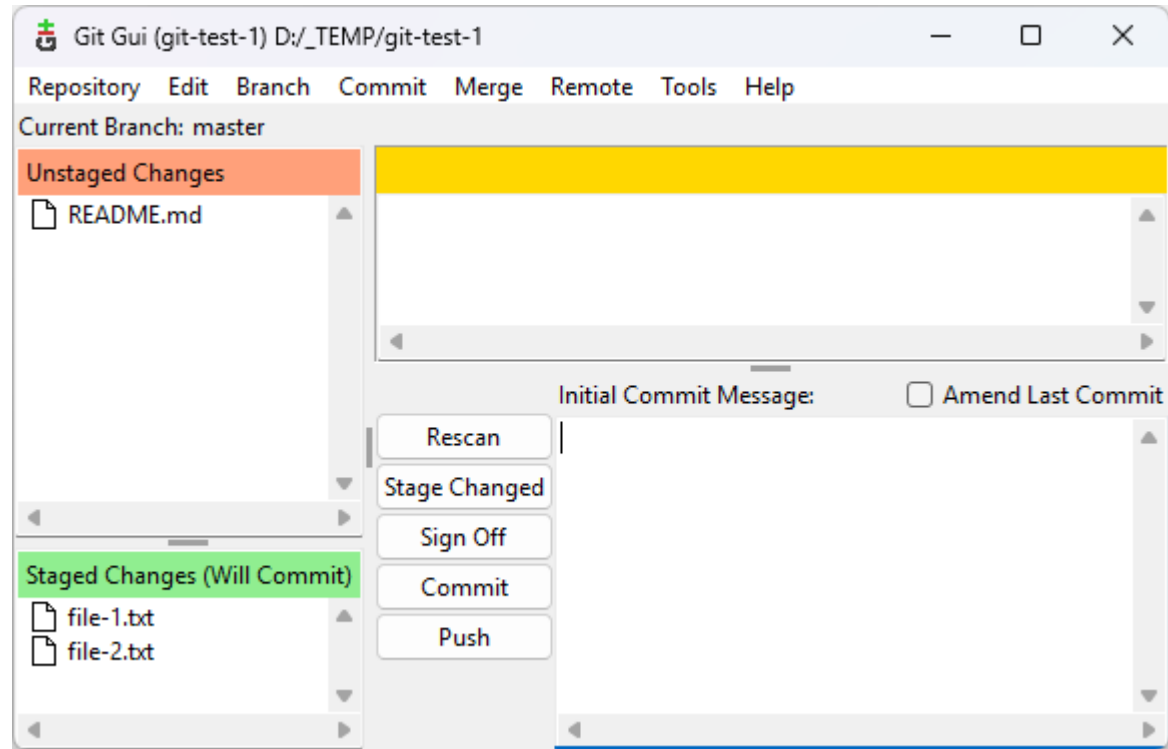
So, CLI is professional's recommendation

CLI – Git Bash

A screenshot of a Git Bash terminal window. The title bar at the top reads "MINGW64:/d/_TEMP/git-test-1" and includes standard window control buttons (minimize, maximize, close). The terminal content shows the prompt "Daryoosh@MyASUS MINGW64 /d/_TEMP/git-test-1 (master)" followed by a dollar sign "\$" on the next line, indicating the shell is ready for input. The terminal background is dark, and the text is in a light color.

```
MINGW64:/d/_TEMP/git-test-1
Daryoosh@MyASUS MINGW64 /d/_TEMP/git-test-1 (master)
$
```

GUI – Git GUI



Glossary

- Working tree/directory
- Staging area (index)
- Repository
 - Local repo (.git directory)
 - For individual/solo working
 - Disconnected from server
 - Remote repo
 - For teamwork collaboration
 - Safety backup

The four states

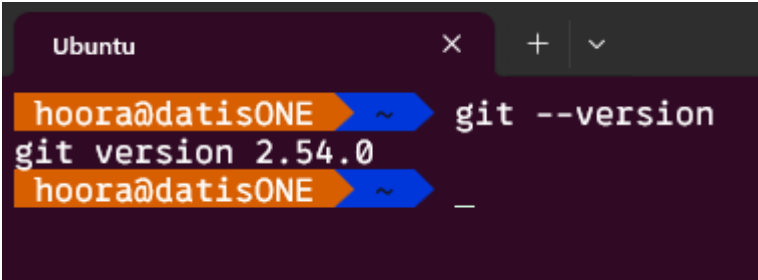
	File state	Location
1	Untracked	Project folder
2	Modified	Working tree
3	Staged	Staging area
4	Committed	Local repository

Git Installation

- Checking if Git is installed or not

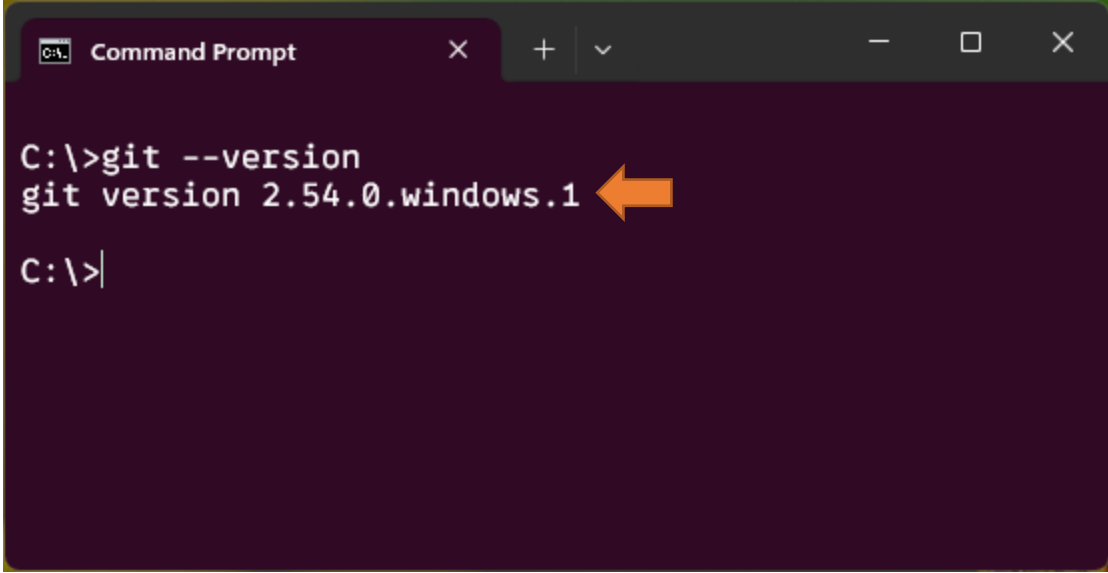
```
git --version
```

```
git -v
```



A terminal window titled 'Ubuntu' showing a user named 'hoora@datisONE' running the command 'git --version'. The output is 'git version 2.54.0'. An orange arrow points to the first prompt line.

```
hoora@datisONE ~$ git --version
git version 2.54.0
hoora@datisONE ~$
```

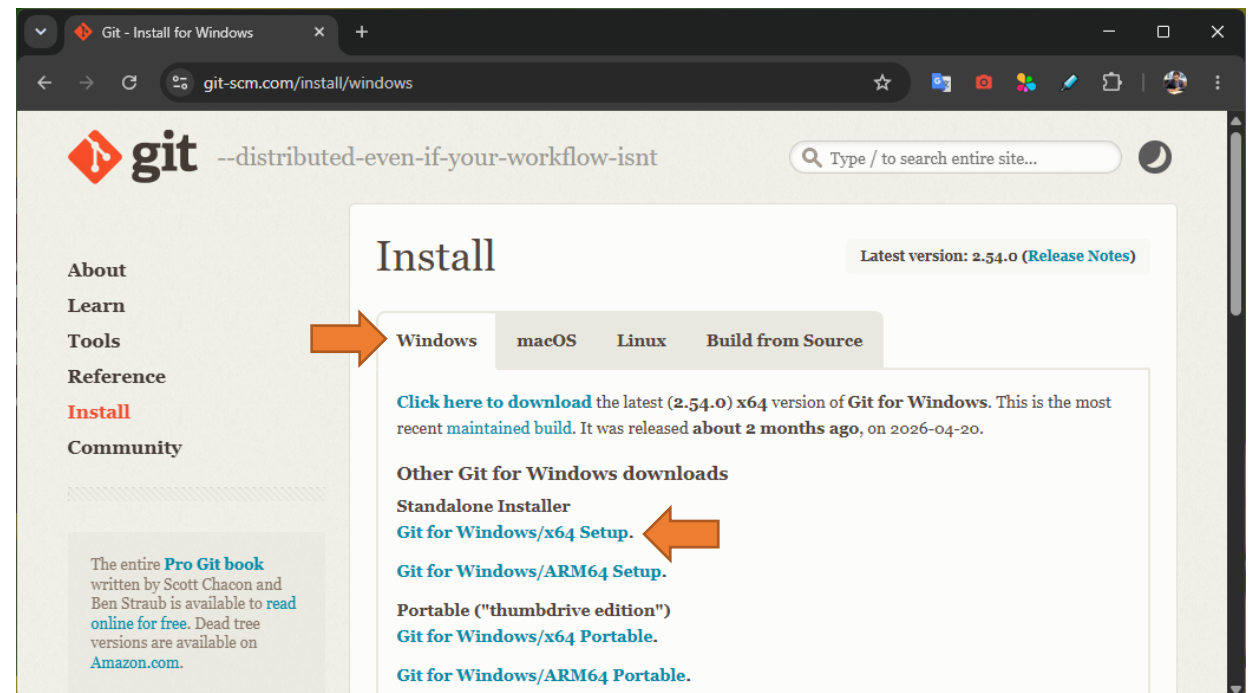


A Windows Command Prompt window showing the command 'git --version' being executed in the C:\ directory. The output is 'git version 2.54.0.windows.1'. An orange arrow points to the output text.

```
C:\>git --version
git version 2.54.0.windows.1
C:\>
```


Git Installation

- Visit the <https://git-scm.com/install/>
- Download the latest version and Install or update to new version.



Git Installation

- Let's try

Git Installation

- On windows:

```
winget install --id Git.Git -e --source winget
```

```
git update-git-for-windows
```

- Install any other GUI clients (GitHub Desktop, SourceTree, Git Tortoise, ...)
- Integration with your IDE

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فصل دوم

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Previously on Git!

- What is Git or is not?
- Who needs Git and why?
- What means of Git concepts?
- How to setting Git up?

Configurations

- git config

```
git config --list
```

- Config files

- [path]/etc/gitconfig (System-level configurations)

```
git config --system <config-item>
```

- ~/.gitconfig or ~/.config/git/config (User-level configurations)

```
git config --global <config-item>
```

- .git/config (project-level configurations)

```
git config --local <config-item>
```

```
or, git config <config-item>
```

```
git config --list --show-origin
```

Local
↓
Global
↓
system

Configurations

- Initialize configurations

- Introduce yourself

- `git config --global user.name "User Name"`
 - `git config --global user.email user@email.com`

Tricks:

- Linux

- `git config --global user.name $USERNAME`

- Windows

- `git config --global user.name $($env:USERNAME)`
 - `git config --global user.email $($env:USERNAME)@$( $env:USERDOMAIN) .com`

Configurations

- Optional configurations
 - Choose your editor
 - https://git-scm.com/book/en/v2/Appendix-C:-Git-Commands-Setup-and-Config#ch_core_editor
 - main or master
 - `git config --system init.defaultBranch main`
 - CRLF or LF
 - `git config --system core.autocrlf [true | false | input]`
 - In Behsazan
 - `git config --system http.sslVerify false`

For more: <https://git-scm.com/docs/git-config>

Getting started

- Create a local repository

- `cd <project-folder>`
- `git init`

- Linking the local repository to a remote repository

- `git remote add origin <url>`
- `git remote -v`
- `git push --set-upstream origin main`

- Clone a remote repository

- `git clone <url>`

Basics

- status
- add
- commit
- diff
- rm
- mv
- reset
- restore
- push
- pull
- fetch

.gitignore

ignore all .a files

**.a*

but do track lib.a, even though you're ignoring .a files above

!lib.a

only ignore the TODO file in the current directory, not subdir/TODO
/TODO

ignore all files in any directory named build

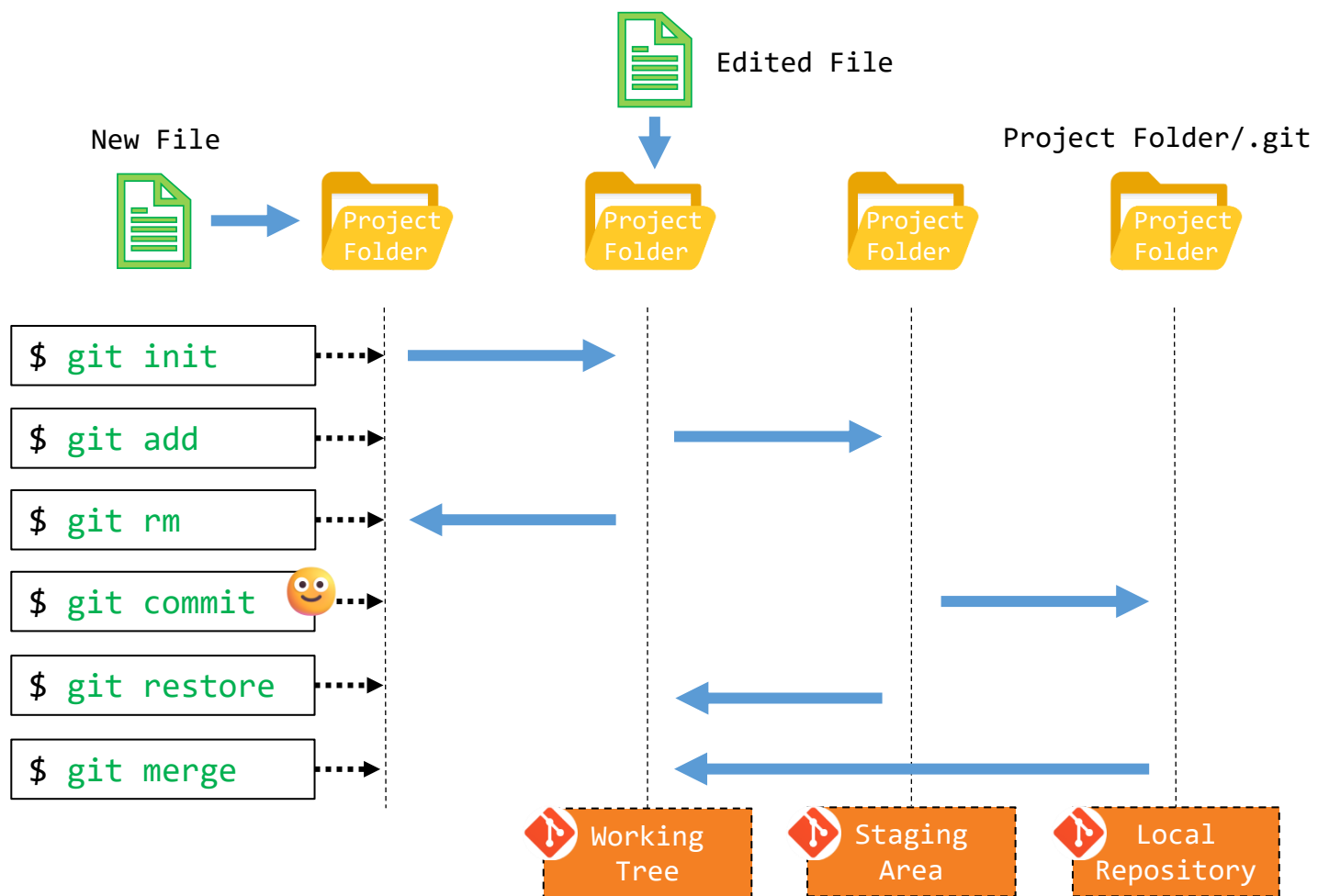
build/

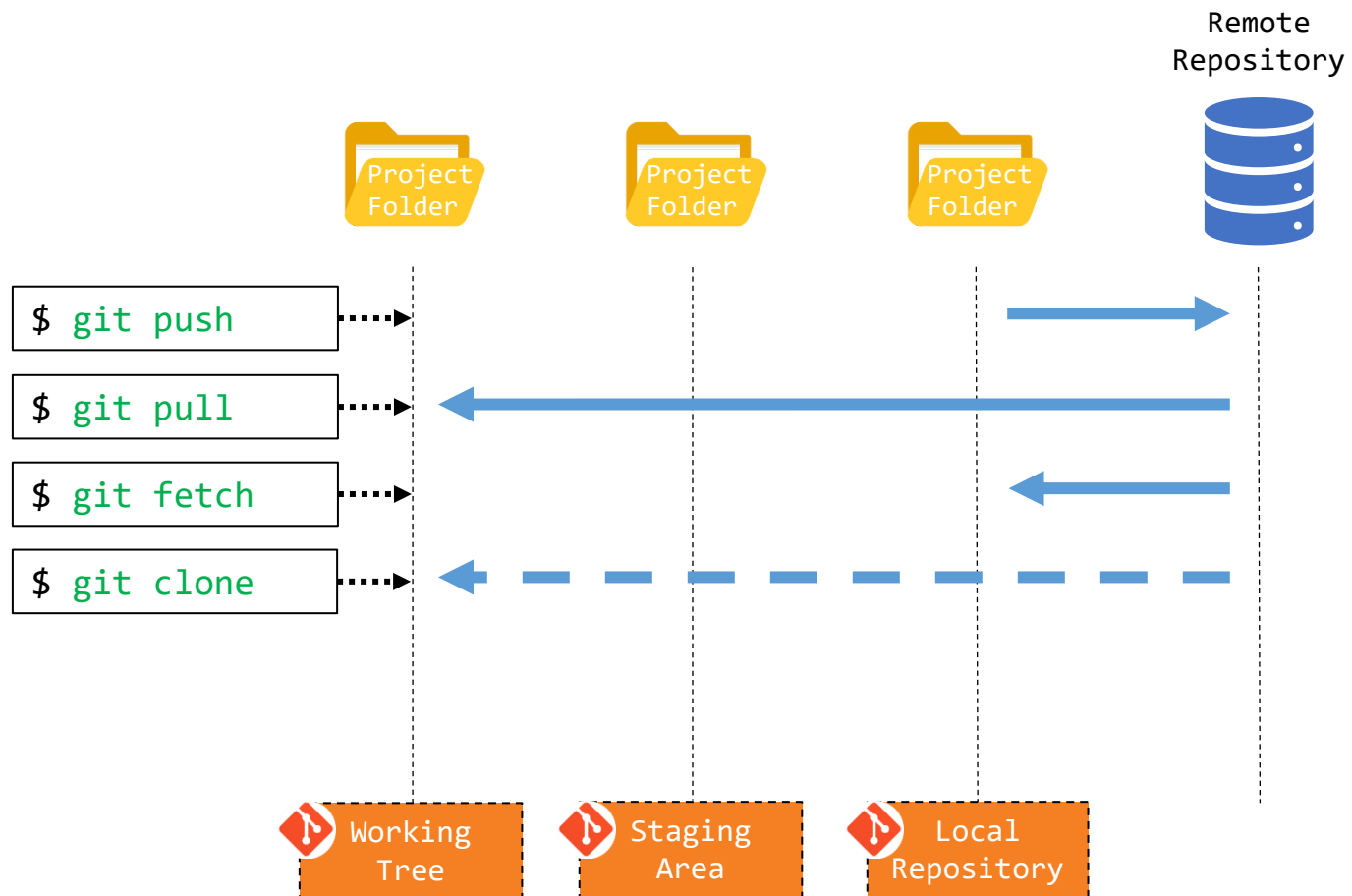
ignore doc/notes.txt, but not doc/server/arch.txt

doc/.txt*

ignore all .pdf files in the doc/ directory and any of its subdirectories

*doc/**/*.pdf*

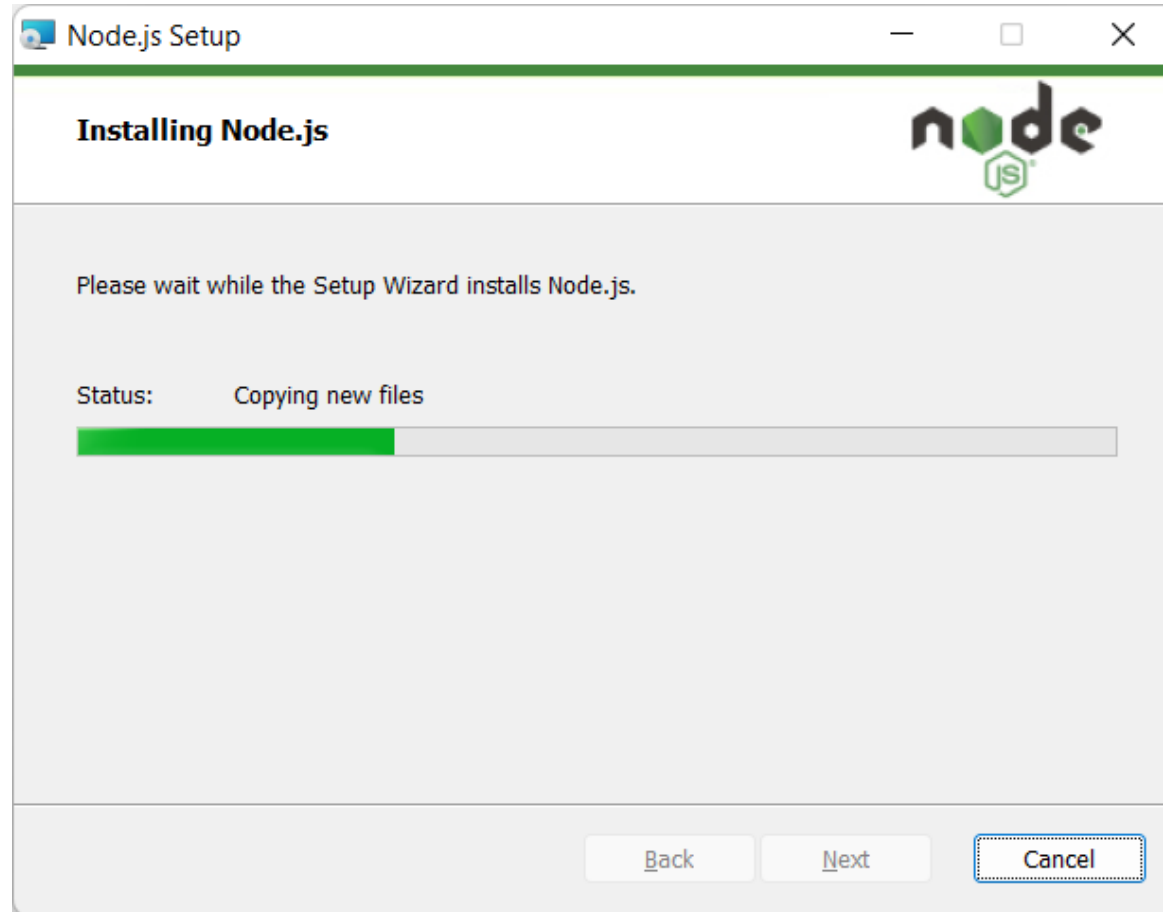




Test scenario

- Create an empty GitHub repository
 - Scenario 1:
 - Linking project folder to remote repository without set default branch.
 - Scenario 2:
 - Linking project folder to remote repository with set default branch.
 - Scenario 3:
 - Linking project folder to remote repository with rename default branch.
- Create a GitHub repository with README file
 - Scenario 4:
 - Linking project folder to remote repository with set default branch.

Node.js-Setup (Continue)



Ok, Ready to start...

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جلسه سوم

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Previously on React!

- Why Node.js?
- How to installing Node.js?
- How to setting up Npm?

We need to,

- Creating a SPA (Single Page App) ← **create-react-app** ←
- Creating a SSR (Server Side Renderer)
- Creating a SCO (Static Content Oriented)
- Creating a Component Library or more...

React Hello World!

www.hoora.info

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Tips and Tricks

- Using the following command to checking version
 > `create-react-app --version`
- Using the following command to uninstall
 > `npm uninstall --location=global create-react-app`
- Reinstall for update

Tips and Tricks

- It's prefer to open windows terminal (Cmd / PowerShell) as Administrator
- Try to use PowerShell instate of Cmd

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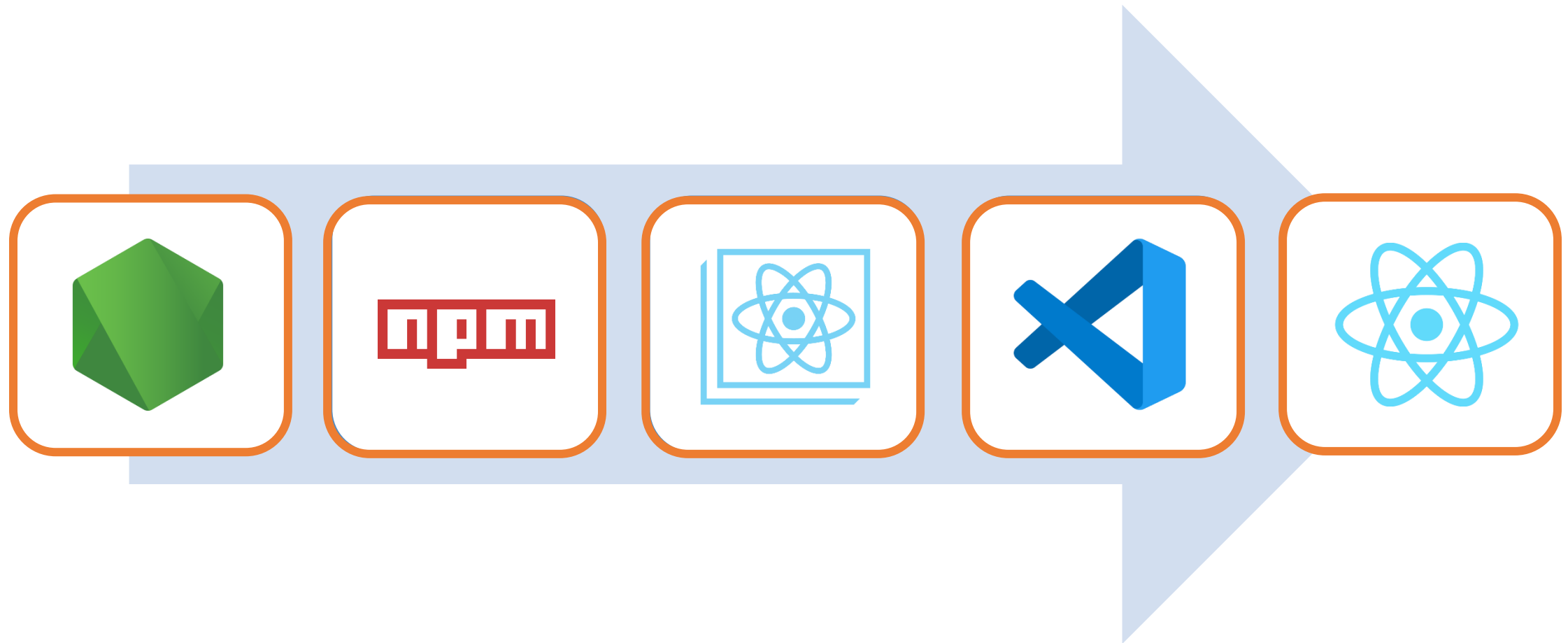


جلسه چهارم

انتشار این محتوای آموزشی با ذکر منبع بلامانع می باشد

Previously on React!

- What is create-react-app?
- How to create React app?
- How to start React app?



Magic of WinGet!

- Create an auto installer batch file!
 1. `winget install -i OpenJS.NodeJS.LTS`
 2. `npm install -g npm@latest`
 3. `npm install -g create-react-app`
 4. `winget install -i microsoft.visualstudiocode`

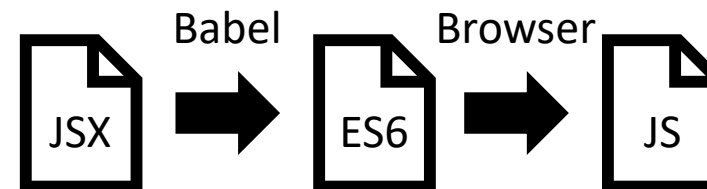


<https://github.com/daryoosh-hoora>

Repositories: **Learning_React.js**

React Glossary

- Component, is a box of UI!
 - Types:
 - Class Component
 - Functional Component
 - Lifecycle:
 - Mount
 - Update
 - Unmount
- DOM (Document Object Model)
- Virtual DOM, is a virtual DOM for speed up
- Render, is about drawing web page
- JSX, is a combination of HTML into JavaScript
- ES6, is ECMAScript
- Babel, is a JavaScript compiler (transpiler)
- Webpack, is a bundling module



Go Live!

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